

Auden Young

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Education

University of California, Berkeley

Berkeley, CA

Engineering Physics B.S. & Math Minor, 3.70 GPA

August 2021–August 2025

- Lab projects on Josephson junctions, violating Bell's inequality, Chladni patterns, optical single qubit gates.
- Graduate solid state physics for devices; advanced lab techniques I & II; quantum field theory, continuum mechanics.
- Functional analysis reading course, complex analysis, probability & random processes, abstract algebra.
- Machine learning, microfabrication, structuring & interpreting computer programs, designing information systems & devices II (circuit analysis & control theory; constructed voice controlled robot car).
- Member of Epsilon Pi (Engineering Physics Honor Society).

Research Experience

University of California, Berkeley

Berkeley, CA

R&D Technician, Physics Advanced Instructional Lab

September 2025–Present

- Hired on as full time technician due to quality of work as Segré intern.
- Assisted with repair of magneto-optical trap experiment through laser replacement, extensive optical alignment, refinement of laser locking strategy. Experiment had not been operational in 8 years.
- Assisted with repair of optical tweezer experiment through optical alignment, rewriting software (MATLAB and Python), DAQ replacement, manual updates. Experiment had not been operational in several years.
- Constructed new muon lifetime experiment station (assembled equipment, performed tests) and assisted with assembly of new solar spectroscopy experiment to allow lab to support twice as many students as in previous semesters.
- Mentored and assisted 10 students in 5 different extension projects; this led to the repair of two experiments (beta ray spectroscopy and laser spectroscopy) that had not been operational in over 10 years, creation of 2 brand new experiments (muon tomography and a new optical pumping apparatus), and improvements to a hall effect in plasma experiment.
- Led first lab cleanup in 8 years; made space for four new experiments and re-organized spare parts for easier instructor and technician access.
- Led maintenance of vacuum pumps which had not been done for 8 years; flushed oil, changed filters, cleaned pumps, baked out foreline traps. Rough pumps all had order of magnitude better vacuum after maintenance.
- Helped transition lab away from Windows 7 to Windows 11; achieved 90% of lab machines upgraded; this required rewriting software (MATLAB, LabVIEW, C#, Python), upgrading equipment, revising manuals, and replacing DAQs in order to make experiments functional with new computers. In process wrote Python package for interfacing with DAQ used throughout lab and implemented version control for all lab software.
- Led improvement of laser safety protocols after two near misses by instituting staff safety goggle rack, interlock key checkout, and student goggle checks with staff before laser turned on. No further near misses occurred post-adjustments.
- Supported instructional staff with daily experiment repairs/maintenance, questions on equipment operation and non-standard problems, manual updates and documentation, etc.
- As result of work, for first time in over a decade, all original experiments in lab operational and 4 new experiments created.

University of California, Berkeley*Emilio Segré Intern, Physics Advanced Instructional Lab***Berkeley, CA***June 2025–August 2025*

- Repaired atomic force microscope experiment and made multiple improvements, including working on adapting instrument for electric force microscopy. Designed new supplementary experiments for students.
- Repaired gamma ray spectroscopy experiment, a process which included upgrading DAQ, rewriting software, and rewriting manual. Resolution of spectrum increased by order of magnitude after repairs and improvements.
- Handled laser and radiation safety inspections.
- Rewrote data acquisition lab for advanced electronics lab course, improved accessibility for students with less programming background and improved code clarity.
- Diagnosed issues, performed electronics repair, software debugging (LabVIEW and Python), optical alignment, and updated documentation. Supported students and instructors in lab.
- Rewrote lab manuals for clarity and future repairs.

Lawrence Berkeley National Laboratory*Student Researcher, Quantum Sensing for Geosciences Group***Berkeley, CA***January 2024–January 2026*

- Worked on quantum sensing of magnetic fields using nitrogen vacancy centers in diamond under Dr. Zhao Hao.
- Designed sample stage for quantum sensing microscope and magnetometer from scratch. Automated stage so that experiments could be run remotely.
- Simulated and designed flux concentrators using FEMM to enhance sensitivity of magnetometer (paper in preparation).
- Wrote code to control hardware and perform automatic statistical analysis, data smoothing using Python.
- Helped design antennas for coherent spin control and PCB for antenna control. Performed E&M simulation to predict performance using openEMS and Python, collected experimental data on performance which showed improved signal to noise ratio (paper in preparation).
- Trained on equipment in the Marvell Nanofabrication Lab to fabricate antennas, including photolithography, chemical benchwork, metal evaporation, descum, liftoff, wirebonding, and measurement tools.
- Assisted in development of high-yield (>90%) process for antenna manufacture and produced several dozen antennas on glass and sapphire substrates (paper in preparation).

University of California, Berkeley*Student Researcher, Quantum Devices Group***Berkeley, CA***November 2021–April 2023*

- Worked on experimental optical quantum computing related to color centers in silicon with Professor Alp Sipahigil. Did part design, rapid prototyping, PCB design, device characterization, and other jobs for group.
- Prior quantum computing work includes developing open source simulator of ideal quantum computer, acceptance to Southern California Junior Academy of Sciences program based on research proposal, writing and defending paper on classical simulation of quantum computers.

Space Sciences Laboratory*Student Researcher, Compton Spectrometer & Imager (COSI)***Berkeley, CA***September 2021–December 2021*

- Accepted through Berkeley's Undergraduate Research Apprentice Program to work with Dr. Andreas Zoglauer.
- Estimated non-measured energy in pair production reactions using 3D CNNs implemented in Keras.

WET Design*Engineering Intern, Engineering & Product Design***Burbank, CA***June 2018–September 2018*

- Developed, maintained, and documented analog computer simulator and testbed.
- Developed new numerical integration technique and co-authored internal paper. Implemented multiple numerical analysis algorithms and simulated chaotic dynamical systems.
- Collaborated with team on product design, prototyping. Took classes on finite element analysis, sheet metal modeling.

Publications

Journal articles

Young, A., Morales De Leon, K., Lahaderne, N., Mogan, S., Kaita Martin, J., Fleming, X., Liu, W., Hao, Z. *Coherent control of nitrogen vacancy centers in diamond using broadband micro-coil antennas*. (In preparation.)

Young, A., Hao, Z. *Increased magnetometer sensitivity using planar flux concentrators*. (In preparation.)

Talks

Hao, Z. (presenter), Singh, N., Fleming, X., Vale, J., Johnson, H., Morales De Leon, K., Young, A., Suwandi, K., Mogan, S., Min, Y. (2026.) *From bulk to nanopore: exploring the physicochemical causes of unconventional behavior of confined water in mineral systems*. Goldschmidt Conference.

Young, A. (presenter), Morales De Leon, K., Lahaderne, N., Mogan, S., Kaita Martin, J., Fleming, X., Liu, W., Hao, Z. (2025.) *Coherent control of nitrogen vacancy centers in diamond using broadband micro-coil antennas*. APS Global Physics Summit.

Young, A. (2025.) *Nitrogen vacancy centers for quantum sensing*. Society for Physics Students Undergraduate Seminar.

Leadership and Service

Lawrence Hall of Science Teen Research Program

Volunteer

June 2025 – August 2025

- Gave lab tours and demonstrated experiments including superconducting magnets, atomic force microscopy, and Zeeman splitting for high school and middle school students.
- Answered middle and high school students' questions about college and physics.

Introduction to Problem Solving in Physics

Co-instructor, DeCal Facilitator

January 2023–May 2025

- Co-founded and co-taught student run class (DeCal) meant to help students adjust to Berkeley Physics. Course served over a hundred students over 2.5 years.
- Taught students introductory mechanics, problem solving strategies, and college study strategies.
- Developed bulk of curriculum. Wrote course materials including worksheets and homeworks. Designed final project. Graded and handled logistics.
- Connected students with campus resources and advocated for struggling students.
- Met with professors for introductory mechanics classes to tailor course to meet student needs.
- Many students said course encouraged them to continue in physics.

Teaching Experience

University of California, Berkeley Physics Department

Berkeley, CA

Physics 7B TA

January 2025–June 2025

- Taught two sections of introductory thermodynamics and electromagnetism under Professor Catherine Bordel, including labs and discussion.
- Wrote worksheets, set up and debugged labs, graded exams, hosted review sessions, connected students with campus resources.
- Instructor efficacy score 6.63/7 (weighted average); department average 5.88/7

University of California, Berkeley Math Department

Berkeley, CA

Math 1 TA

August 2024–December 2024 & August 2023–December 2023

- Taught four sections of freshmen and sophomores learning precalculus under Professor Alexander Paulin.
- Designed and wrote review worksheets, graded homework and exams, guided students in the transition to college, connected students to campus resources.
- Instructor efficacy score 6.5/7 (weighted average); department average 5.95/7.

University of California, Berkeley Student Learning Center

Berkeley, CA

Math Summer Bridge Tutor

June 2023–August 2023 & June 2022–August 2022

- Tutored college algebra and precalculus for rising freshmen from underserved high schools.
- Recommended by program coordinator for TA position in Math Department based on job performance and student feedback.

Awards & Honors

Physics Department Extra Mile Award

University of California, Berkeley

May 2026

- Awarded the Physics Department Extra Mile Award for work improving the Advanced Instructional Lab.

Outstanding Graduate Student Instructor Award

University of California, Berkeley

January 2026

- Awarded the Outstanding Graduate Student Instructor award for teaching in Physics 7B, in recognition of exceptional achievements as a teacher.

Emilio Segré Intern

University of California, Berkeley

June 2025

- Awarded the Emilio Segré internship, for undergraduate and graduate students to improve the advanced physics instructional labs via the design of new experiments and the repair and improvement of existing ones.

Engineering Physics Department Citation Finalist

University of California, Berkeley

May 2025

- Finalist for department citation, awarded to student in department that exhibited strong academic performance, research ability, and community participation.

Maria Goeppert Mayer Fellowship

University of California, San Diego

February 2025

- Fellowship awarded to one student annually for “exceptional academics, research, and strong community engagement in the area of diversity, equity, and inclusion.”
- Fellowship canceled due to funding cuts.

Skills

Programming languages: Python (including NumPy, SciPy, Matplotlib, Pandas, Tensorflow, PyVISA, ScopeFoundry), LabVIEW, Mathematica, MATLAB, RISC-V, HTML/CSS.

Software: LaTeX, Git, Autodesk Inventor, Solidworks, Fusion 360 (for CAD, PCB design, finite element analysis), FEMM (magnetostatics simulations), LTspice, Gwyddion, Google Drive, MS Office, Linux, Windows.

Technical skills: Microfabrication (photolithography, thin film deposition, liftoff processes, wire-bonding, and related tools), atomic force microscopy, PCB design, soldering, 3D printing, technical drawing including GD&T, data pipeline design (acquisition, signal processing, statistical analysis), neural network training and design, hardware-software integration, finite element analysis, optical alignment (certified Lawrence Berkeley National Lab training course), vacuum system maintenance, basic power tool usage, Arduino, Raspberry Pi.

Other proficiencies: Technical writing/documentation, educational writing, public science communication, tutoring and teaching, curriculum design, component sourcing.